

Plasmid preparation by alkaline lysis

Risk assessment

▷ Wear gloves, safety glasses, lab coat



Reviewed: May 30, 2023

This is a bench card. Full protocol available online.

>> Alkaline lysis

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| ☐ R0006 Buffer P1 (Resuspension buffer), 250 mL | ☐ R0009 Buffer N3 (Neutralization buffer), 250 mL |
| ☐ R0007 Buffer P2 (Lysis buffer), 250 mL | |

- (1.) Grow bacteria in 1–5 mL LB (Miller) overnight to stationary phase. Collect the cells by centrifugation.
- (2.) Resuspend the pellet in 150 µL Buffer P1.
- (3.) Add 250 µL Buffer P2. Mix by gently inverting the tube 6–8 times. Incubate at room temperature for no more than 5 min or until the lysate appears clear.

Critical: Do not vortex or pipette to avoid shearing chromosomal DNA.

- (4.) Add 350 µL Buffer N3. Mix gently by inversion 6–8 times.

Critical: Ensure complete neutralization to precipitate protein and chromosomal DNA. If Buffer P2 contains thymolphthalein, mix until the solution is colorless.

- (5.) Centrifuge at $17\,000 \times g$ for 3 min to pellet precipitated material. Repeat if the lysate remains turbid.
- (6.) Transfer the clear supernatant to a fresh tube for purification. Avoid any carryover of precipitate.

A > Purification of plasmid DNA over a silica matrix

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| ☐ R0010 Buffer PB (Binding buffer), 250 mL | ☐ R0056 5 mM Tris hydrochloride |
| ☐ R0011 Buffer PE (Wash buffer), 250 mL | |

- (1.) Apply up to 700 µL of supernatant to a silica spin column. Centrifuge at $11\,000 \times g$ for 1 min, or use a vacuum manifold.
- (2.) *Optional:* Wash the silica matrix with 450 µL Buffer PB.
- (3.) Wash with 650 µL Buffer PE.
- (4.) Dry the silica matrix by centrifugation at $11\,000 \times g$ for 2 min.
- (5.) Place the spin column into a clean microcentrifuge tube.
- (6.) Apply 25–55 µL 5 mM Tris pH 8.0 to the membrane. Incubate for 1 min, then elute the plasmid by centrifugation at $11\,000 \times g$ for 1 min.
- (7.) *Optional:* Repeat elution using the same or half the volume to improve yield. Pool eluates if desired.

B > Purification by alcohol precipitation

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| ☐ Isopropyl alcohol | ☐ 5 mM Tris |
| ☐ Buffer PE (Wash buffer) | |

- (1.) Transfer up to 700 μ L of supernatant to a clean microcentrifuge tube.
- (2.) Add an equal volume of isopropanol to precipitate the DNA. Vortex to mix thoroughly.
- (3.) Centrifuge at $11\,000 \times g$ for 5 min. Discard the supernatant and briefly spin to remove residual solvent.  The pellet will appear nearly transparent.
- (4.) Wash with 650 μ L Buffer PE or freshly prepared 70% ethanol without disturbing the pellet.
- (5.) Air-dry the pellet for 2–5 min, or dry in a vacuum centrifuge for 1 min.
- (6.) Resuspend in 25–55 μ L 5 mM Tris pH 8.0 or water.

C > Purification of plasmid DNA by anion-exchange chromatography

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| ☐  R0008 Buffer P3 (Neutralization buffer), 250 mL | ☐  R0198 Buffer QC (Wash buffer), 250 mL |
| ☐  R0197 Buffer QBT (Equilibration buffer), 250 mL | ☐  R0199 Buffer QF (Elution buffer), 250 mL |

- (1.) Equilibrate the anion-exchange column with the recommended volume of Buffer QBT and allow it to drain by gravity flow. 
- (2.) Apply the P3-neutralized, cleared lysate to the equilibrated column and let it pass through by gravity.

Critical: The column must not run dry between steps; if flow stalls, refill with Buffer QC and continue. 
- (3.) Wash the column twice with the recommended volume of Buffer QC.
- (4.) Elute the plasmid DNA with the recommended volume of Buffer QF into a clean tube. 
- (5.) Precipitate the DNA by adding 0.7 vol isopropanol at room temperature, wash, and resuspend as described in *Alternative B*.

 Recipe (available online)  Troubleshooting (available online)  Notes (available online)

